Korea Technology and Information Promotion Agency for SMEs

# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

| Proposer | Name | Dr. Mateusz Balcerzak | Email | mateusz.balcerzak@ifam-dd.fraunhofer.de |
|---|---|---|---|---|
| Institute Name | Fraunhofer IFAM Dresden | | | |
| Address | Winterbergstraße 28, 01277 Dresden, Germany | | | |

## 2. Technology Overview

| | | | | |
|---|---|---|---|---|
| Title of Proposed Technology | *System to separate and recover fuel for nuclear fusion* | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power (√), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | *Energy & Resources* | | *Energy & Resources* | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | *Nuclear Power* | *Nuclear Fuel and Component Material* | *New & Renewable Energy* | *Hydrogen* |
| Concept of Proposed Technology | *The proposed R&D project focuses on the development of a dedicated system for separation and recovery of future fusion fuel - tritium (following the breeding process and recovery from unreacted fusion fuel streams). The core concept is to design, build, and demonstrate a prototype tritium separation and recovery unit that can be integrated into future fusion reactor fuel cycles, including pilot and DEMO-class fusion facilities (alternatively the system can be used to separate tritium in facilities producing tritium for alternative applications - power self-illuminating exit signs, watch dials, and firearm sights). The system will be optimized for application in breeding blanket processing and reactor exhaust handling, supporting closed fuel-cycle operation.*<br><br>*The technology will be based on metal-hydride systems that selectively capture, separate, store, and release hydrogen isotopes under controlled conditions. Metal hydrides provide high isotope selectivity, compact design, and inherently safe low-pressure operation, making them well suited for fusion environments where safety, containment, and accountability of tritium are critical. The project* | | | |

| | |
|---|---|
| | *will develop modular, tritium-compatible hydride-based recovery units with integrated thermal management, monitoring, and control functions.*<br><br>*The project will progress from fusion-specific system requirement definition through detailed design, manufacturing, and validation of pilot-scale modules. Testing will be conducted using protium and deuterium molecules, leading to system-level demonstration in relevant fusion fuel-cycle test loops. The final outcome will be a product-oriented prototype system advanced to TRL 6–7, bridging the gap between laboratory-scale concepts and deployable fusion reactor components.* |
| **Rationale for Joint R&D Project** | *Efficient separation, recovery, and recycling of tritium is a critical enabler for sustainable nuclear fusion power. Future fusion reactors must achieve fuel self-sufficiency and minimize tritium losses, yet current fusion fuel-cycle technologies rely on complex, centralized systems that are not optimized for compact, modular, and highly selective recovery of tritium from breeding and exhaust streams. This creates a clear technological gap as fusion programs move toward pilot and DEMO-scale operation.*<br><br>*Metal-hydride-based technologies offer a promising alternative due to their high isotope selectivity, solid-state operation, and intrinsic safety advantages. However, their application to fusion fuel cycles requires dedicated R&D addressing tritium compatibility, radiation effects, nuclear-grade safety integration, and regulatory compliance. Such development is technically complex, high risk, and requires specialized facilities and expertise, making it difficult to pursue without public support.*<br><br>*A joint R&D project between Fraunhofer IFAM and a Korean SME enables the combination of advanced IFAM expertise and IP in metal hydrides (also in the field of hydrogen isotope separation) and hydrogen isotope handling with Korean fusion system requirements, industrial capabilities, and regulatory frameworks (1-3). This collaboration supports alignment with national fusion programs such as KSTAR and future DEMO concepts, accelerates technology maturation, and positions Korean industry to play a leading role in the emerging global fusion fuel-cycle supply chain.* |
| **Key Words** | 1. *Tritium*<br>2. *Separation*<br>3. *Recovery*<br>4. *Fusion* |

## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level (3-4) | End of the Project | Level (6-7) |
|---|---|---|---|---|
| **R&D Objectives** | *Primary objective of the project is to develop and demonstrate a prototype system for tritium separation and recovery suitable for integration into fusion fuel cycles. The project aims to deliver product-oriented, metal-hydride-based recovery modules with validated performance, safety, and maintainability under fusion-relevant operating conditions.*<br><br>*Key objectives include achieving efficient hydrogen isotope(in the future from recovery from breeding and exhaust streams), demonstrating reliable isotope-selective separation, and enabling controlled short and medium term storage and release of hydrogen isotope for fuel recycling. The project also targets the development of tritium-compatible designs that meet nuclear safety and regulatory requirements and support remote handling and long-term operation.*<br><br>*The project will culminate in a system-level prototype validated in relevant test environments, supported by technical documentation, safety data, and an exploitation roadmap, enabling advancement from an initial TRL of approximately 3–4 to a final TRL of 6–7 and supporting future commercialization and deployment in fusion facilities.* | | | |
| **R&D Target Specifications** | *The developed tritium separation and recovery system will target a tritium recovery efficiency of at least 95 percent, with an allowable tritium loss below 5 percent of processed inventory under steady-state operation. The system will be designed to handle fusion-relevant gas streams with throughputs corresponding to pilot and DEMO-class fuel-cycle requirements, with modular scalability for higher capacities.*<br><br>*The metal-hydride-based recovery modules shall demonstrate selective hydrogen isotope separation with product purity exceeding 99.9 percent, while operating at low to moderate pressures below 10 bar and temperatures typically below 150 °C (depending on the final concept higher temperatures are also possible). The system will support repeated absorption and desorption cycles with less than 10 percent performance degradation over cycling.*<br><br>*From a safety and operability perspective, the system can incorporate suitable confinement barriers, continuous hydrogen isotope monitoring, and fail-safe thermal control. All components will be compatible with tritium exposure, radiation environments, and nuclear-grade materials standards. The final prototype will meet functional and safety requirements necessary for validation at TRL 6–7 in fusion fuel-cycle test loops.* | | | |
| **R&D Plan** | *Overall approach and timeline (36 months total)*<br><br>*The project is structured into two sequential phases, progressing from defining of functional and operational requirements and preliminary system concept definition to manufacturing and demonstrating of a pilot-scale separation unit.* | | | |

| | |
|---|---|
| | ***Phase I Pre-study and Concept Definition (6 months)***<br>• *Definition of functional and operational requirements for tritium separation and recovery aligned with Korean fusion programs and DEMO concepts*<br>• *Specification of performance targets for recovery efficiency, throughput, purity, and allowable tritium losses*<br>• *Review of applicable Korean nuclear safety, tritium handling, and licensing requirements*<br>• *Selection of suitable metal-hydride materials and preliminary system architecture*<br>• *Development of a preliminary system concept and exploitation strategy*<br><br>*Milestone Phase I: Approved fusion-relevant system concept with defined requirements and go decision for prototype development*<br><br>***Phase II Main R&D and Demonstration (30 months)***<br>• *Design and optimization of metal-hydride-based tritium recovery modules including thermal management and safety systems*<br>• *Manufacturing of pilot-scale, tritium-compatible modules and system integration*<br>• *Testing with hydrogen and deuterium isotopes (potentially followed by tritium testing in licensed facilities – if applicable)*<br>• *Safety, reliability, and maintainability assessment under representative operating conditions*<br>• *Demonstration in fusion fuel-cycle test loops relevant to KSTAR and future DEMO systems*<br><br>*Milestone Phase II: Validated prototype tritium separation and recovery system achieving TRL 6–7* |

## 4. R&R (Roles and Responsibilities)

| | |
|---|---|
| **Expected R&R of Research Institute** | *Fraunhofer IFAM will provide scientific and technological co-leadership for the project, focusing on metal-hydride materials, hydrogen isotope behavior, and system-level integration. The institute will lead material selection, hydride-bed design, and performance validation, including advanced in-situ and operando studies where required. Fraunhofer IFAM will contribute expertise in tritium-compatible system design, safety concepts, and fuel-cycle integration. The institute will also support TRL assessment, safety documentation, and knowledge transfer to the industrial partner.* |
| **Expected R&R of Korean SME Partner** | *The Korean SME will lead application definition, system engineering, manufacturing, and industrialization aspects of the project. Responsibilities include fabrication of pilot-scale modules, integration into fusion-relevant test environments, and support for long-term testing and maintainability assessment. The SME will also lead commercialization planning, regulatory alignment, and preparation for future deployment within Korean fusion programs. Close cooperation with Fraunhofer IFAM will ensure alignment between advanced materials development and practical fusion system requirements.* |

## 5. Expected Outcomes/Impact

| | |
|---|---|
| **Expected Outcomes and Impact** | *The project will deliver a validated prototype system for tritium separation and recovery, providing a critical building block for closed fusion fuel cycles. Technologically, it will establish a new solution for efficient, safe, and compact tritium recycling, reducing fuel losses and supporting fuel self-sufficiency in future fusion reactors. The prototype and associated data will support licensing, further scale-up, and deployment in pilot and DEMO-class fusion facilities.*<br><br>*From an economic and strategic perspective, the project will position the participating organizations among the first suppliers of specialized fusion fuel-cycle technologies. It is expected to enable early entry into the emerging global fusion component market, with potential to capture a significant share of fuel-cycle systems as fusion facilities are deployed internationally. The project will also strengthen Korea's industrial capabilities in fusion technologies, create high-skilled jobs, and contribute to long-term leadership in the global fusion energy ecosystem.* |

## 6. References

(1) EP3045910 B1 patent, Messeinrichtung und Messverfahren zur Bestimmung der Menge eines in einem Speicher aufgenommenen Gases an einem porösen Speichermaterial

(2) EP 3 118 511 B1 patent, Speicherelement für Gase

(3) DE 10 2020 204 214 B4 patent, Behälter zum Speichern von Wasserstoff

## ☐ Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| Description | | |
|---|---|---|
| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |